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ADJUSTABLE SHOCK ASSEMBLY

Abstract

An adjustable shock assembly is disclosed herein. The adjustable shock assembly includes a damper body having a main chamber with a working fluid therein. An adjustable active valve assembly comprising: a first parallel flow pathway comprising an active valve; and a second parallel flow pathway comprising a firm mode blowoff stack and a firm mode adjuster to adjust a blowoff pressure of the firm mode blowoff stack. A main piston coupled with a piston shaft, and a fluid pathway fluidly coupling the main chamber with the adjustable active valve assembly which is fluidly coupled with a reservoir

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of and claims benefit of U.S. patent application Ser. No. 17/841,450, filed Jun. 15, 2022, entitled “ADJUSTABLE SHOCK ASSEMBLY” by Laird et al., assigned to the assignee of the present application, having Attorney Docket No. FOX-0161US, which is herein incorporated by reference in its entirety. [0002] The application Ser. No. 17/841,450 claims benefit of U.S. Provisional Patent Application Ser. No. 63/211,217, filed Jun. 16, 2021, entitled “METHOD AND APPARATUS FOR AN ADJUSTABLE DAMPER” by Laird et al., assigned to the assignee of the present application, having Attorney Docket No. FOX-0161US.PRO, which is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] Embodiments of the invention generally relate to adjustable shock assemblies.

BACKGROUND

[0004] Shock assemblies (e.g., dampers, shock absorbers, etc.) are used in numerous different vehicles and configurations to absorb some or all of a movement that is received at an unsprung portion of a vehicle before it is transmitted to a suspended portion of the vehicle. For example, when a wheel hits a pothole, the encounter will cause an impact force on the wheel. However, by utilizing suspension components including one or more shock assemblies, the impact force can be significantly reduced or even absorbed completely before it is transmitted to a person on a seat of the vehicle. However, depending upon the terrain being traversed, it can be valuable to be able to change the amount of shock absorption provided by the shock assembly for personal comfort, vehicle performance, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present invention are illustrated by way of example, and not by way of limitation, in the accompanying drawings, wherein:

[0006] FIG. 1A is a perspective view of a bicycle having an active suspension system and an adjustable shock assembly, in accordance with an embodiment.

[0007] FIG. 1B is a schematic side view of a bicycle with the active suspension system and adjustable shock assembly, in accordance with an embodiment.

[0008] FIG. 2 is a perspective view of an adjustable shock assembly, in accordance with an embodiment.

[0009] FIG. 3 is a side view of the upper portion of the adjustable shock assembly including a cutaway view of the damper body and the adjustable rebound valve assembly, shown in accordance with an embodiment.

[0010] FIG. 4A is a side view of the upper portion of the adjustable shock assembly including a cutaway view of the adjustable active valve assembly and the adjustable compression valve assembly, shown in accordance with an embodiment.

[0011] FIG. 4B is a side view of the adjustable shock assembly including a cutaway portion of a reservoir, shown in accordance with one embodiment is shown in accordance with an embodiment. [0012] FIG. 5 is a schematic diagram of the fluid flow in the adjustable shock assembly, shown in accordance with an embodiment.

[0013] The drawings referred to in this description should be understood as not being drawn to scale except if specifically noted.

DESCRIPTION OF EMBODIMENTS

[0014] The detailed description set forth below in connection with the appended drawings is intended as a description of various embodiments of the present invention and is not intended to represent the only embodiments in which the present invention is to be practiced. Each embodiment described in this disclosure is provided merely as an example or illustration of the present invention, and should not necessarily be construed as preferred or advantageous over other embodiments. In some instances, well known methods, procedures, and objects have not been described in detail as not to unnecessarily obscure aspects of the present disclosure.

[0015] In general, a suspension system for a vehicle provides a motion modifiable connection between a portion of the vehicle that is in contact with a surface (e.g., an unsprung portion) and some or all of the rest of the vehicle that is not in contact with the surface (e.g., a suspended portion). For example, the unsprung portion of the vehicle that is in contact with the surface can include one or more wheel(s), skis, tracks, hulls, etc., while some or all of the rest of the vehicle that is not in contact with the surface include suspended portions such as a frame, a seat, handlebars, engines, cranks, etc.

[0016] The suspension system will include one or numerous components which are used to couple the unsprung portion of the vehicle (e.g., wheels, skids, wings, etc.) with the suspended portion of the vehicle (e.g., seat, cockpit, passenger area, cargo area, etc.). Often, the suspension system will include one or more shock assemblies which are used to reduce feedback from the unsprung portion of the vehicle before that feedback is transferred to the suspended portion of the vehicle, as the vehicle traverses an environment. However, the language used by those of ordinary skill in the art to identify a shock assembly used by the suspension system can differ while referring to the same (or similar) types of components. For example, some of those of ordinary skill in the art will refer to the shock assembly as a shock absorber, while others of ordinary skill in the art will refer to the shock assembly as a damper (or damper assembly).

[0017] Described in more detail herein, the adjustable shock assembly often consists of a (damping) piston and piston rod telescopically mounted in a fluid filled cylinder (e.g., a housing). The fluid (e.g., damping fluid, working fluid, etc.) may be, for example, a hydraulic oil, a gas such as nitrogen, air, or the like. In one embodiment, the adjustable shock assembly will include a mechanical spring (e.g., a helically wound spring that surrounds or is mounted in parallel with the body of the adjustable shock assembly). In one embodiment, the adjustable shock assembly will include an air spring. In one embodiment, the adjustable shock assembly will include both a mechanical spring and an air spring.

[0018] As used herein, the terms “down”, “up”, “downward”, “upward”, “lower”, “upper”, and other directional references are relative and are used for reference and identification purposes.

[0019] In its basic form, the suspension is used to increase ride comfort, performance, endurance, component longevity and the like. In general, the force of jarring events, rattles, vibrations, jostles, and the like which are encountered by the portion of the vehicle that is in contact with the surface are reduced or even removed as it transitions through the suspension before reaching suspended portions of the vehicle to include components such as seats, steering wheels/handlebars, pedals/foot pegs, fasteners, drive trains, engines, and the like.

[0020] For example, on a wheeled vehicle, a portion of the wheel (or tire) will be in contact with the surface being traversed (e.g., pavement, dirt, gravel, sand, mud, rocks, etc.) while a shock assembly and/or other suspension system components will be coupled between a wheel retaining

assembly and the suspended portion of the vehicle (often a portion of the vehicle frame and associated systems, the seat, handlebars, pedals, controls, steering wheel, interior, etc.).

[0021] In a snow machine, a portion of the track and/or the skis that will be in contact with the surface being traversed (e.g., snow, ice, etc.) while a shock assembly and/or other suspension components will be coupled between a track retaining assembly (and similarly the skis retaining assembly) and the suspended portion of the vehicle (usually including the engine and associated systems, the seat, handlebars, etc.).

[0022] In a boat or PWC vehicle, a portion of the hull will be in contact with the surface of the water while a shock assembly and/or other suspension components will be coupled between the hull and the suspended portion(s) of the vehicle (such as the seat, the handlebars, a portion of the vehicle frame, and/or the like).

[0023] In an airplane in flight, it is the airframe that is in contact with the surface being traversed (e.g., the air) while a shock assembly and/or other suspension components will be coupled between the airframe and the suspended portion(s) of the vehicle (such as the seats and the like).

[0024] The term ride height refers to a distance between a portion of a vehicle and the surface across which the vehicle is traversing. Often, ride height is based on one or more of a number of different measurements such as, but not limited to, a distance between a part of the vehicle and the ground, a measurement between the top of an unsprung portion of a vehicle and a suspended portion of the vehicle there above, etc. For example, a portion of the wheel(s) (or ski, track, hull, etc.) will be in contact with the surface, while one or more adjustable shock assemblies and/or suspension components maintain the suspended portion of the vehicle a certain height there above.

[0025] The term initial sag settings or “sag” refers to a pre-defined vehicle ride height and suspension geometry based on the initial compression of one or more shock assemblies of the suspension system for a given vehicle when it is within its normal load envelope configuration (e.g., with a rider/driver and any initial load weight). Once the sag is established for a vehicle, it will be the designated ride height of the vehicle, until and unless the sag is changed.

[0026] The initial sag for a vehicle is usually established by the manufacturer. The vehicle sag can then be modified and/or adjusted by an owner, a mechanic, or the like. For example, an owner can modify the sag to designate a new normal ride height based on a vehicle use purpose, load requirements that are different than the factory load configuration, an adjustment modification and/or replacement of one or more of the suspension components, a change in tire size, a performance adjustment, aesthetics, and the like.

[0027] In one embodiment, the initial manufacturer will use sag settings resulting in a pre-established vehicle ride height based on vehicle use, size, passenger capacity, load capacity, and the like. For example, a truck (side-by-side, car, or the like) may have a pre-established sag based on an expected load (e.g., a number of passengers, an expected cargo requirement, etc.).

[0028] Regardless of the vehicle type, once the sag is established, in a static situation the ride height of the expectedly loaded vehicle should be at or about the established sag. When in motion, the ride height will change as the vehicle travels over the surface, and while the suspension system is used to reduce the transference of any input forces received from the surface to the rest of the vehicle it is also used to maintain the vehicle's sag.

[0029] However, when additional weight is added to the vehicle, the suspension and one or more adjustable shock assemblies will be compressed, and the vehicle ride height will be less than the sag.

[0030] Additional information regarding sag and sag setup can be found in U.S. Pat. No. 8,838,335 which is incorporated by reference herein, in its entirety.

[0031] As vehicle utilization scenarios change, one or more shock assemblies of the suspension system can be adjusted for different characteristics based on the use type of the vehicle, terrain, purpose (e.g., rock crawl, normal use, race set-up, etc.), and the like. This modification would result in a modified personal sag setting. For example, a downhill mountain bike rider (motocross

rider, off-road truck driver, side-by-side rider, snow machine racer, etc.) would want a suspension configuration with a large range of motion and aggressive rebound and compression speeds to maintain as much contact as possible between the tires and the ground by absorbing the terrain events such as bumps, ruts, roots, rocks, dips, etc. while reducing the impacts felt at the suspended portion and also have the suspension return to its personal sag setting as quickly as possible in preparation for the next encounter.

[0032] In contrast, a street bike racer (track racing vehicle, boat/PWC racer, etc.) would want a firmer suspension configuration with a very small range of motion to provide feel for the grip of the tire, maintain friction and/or aerodynamic geometries, and the like, in order to obtain the maximum performance from the vehicle.

[0033] In one embodiment, there may be times where changes to a suspension component are desired during a given ride/drive. For example, a bike rider in a sprinting scenario would often want to firm up or possibly even lockout the suspension component to remove the opportunity for rider induced pedal bob. Similarly, a ride/drive from a paved road to an off-road environment (or vice-versa) would also be a time when a change to one or more suspension component settings is valuable.

[0034] With respect to the term lockout, for purposes of the following discussion, lockout refers to the most restricted flow state attainable or desirable. Thus, in one embodiment, lockout refers to a stoppage of all fluid flow through a given fluid path. However, in another embodiment, lockout does not stop all the fluid flow through a given fluid path. For example, a manufactured component may not be able to stop all fluid flow due to tolerances, or a manufacturer (designer, etc.) may not want to stop all fluid flow for reasons such as lubrication, cooling, etc. Similarly, a lockout state could be a “perceived lockout”; that is, the flow area through a flow path of the adjustable shock assembly has been reduced to a minimum size for a given adjustable shock assembly, machine, environment, speed, performance requirement, etc. For example, in one “perceived lockout” most, but not all, of the fluid flow is minimized while in another “perceived lockout” the fluid flow is reduced by only half (or a third, quarter, three-quarters, or the like).

[0035] In the following discussion, the term “active”, as used when referring to a valve or shock assembly component, means adjustable, manipulatable, etc., during typical operation of the valve. For example, an active valve can have its operation changed to thereby alter a corresponding shock assembly characteristic damping from a “soft” setting to a “firm” setting (or a stiffness setting somewhere therebetween) by, for example, adjusting a switch in a passenger compartment of a vehicle. Additionally, it will be understood that in some embodiments, an active valve may also be configured to automatically adjust its operation, and corresponding shock assembly damping characteristics, based upon, for example, operational information pertaining to the vehicle and/or the suspension with which the valve is used.

[0036] Similarly, it will be understood that in some embodiments, an active valve may be configured to automatically adjust its operation, and corresponding shock assembly damping characteristics, based upon received user input settings (e.g., a user-selected “comfort” setting, a user-selected “sport” setting, and the like). In many instances, an “active” valve is adjusted or manipulated electronically (e.g., using a powered solenoid, or the like) to alter the operation or characteristics of a valve and/or other component. As a result, in the field of suspension components and valves, the terms “active”, “electronic”, “electronically controlled”, and the like, are often used interchangeably.

[0037] In the following discussion, the term “manual” as used when referring to a valve or shock assembly component means manually adjustable, physically manipulatable, etc., without requiring disassembly of the valve, damping component, or shock assembly which includes the valve or damping component. In some instances, the manual adjustment or physical manipulation of the valve, damping component, or shock assembly which includes the valve or damping component, occurs when the valve is in use. For example, a manual valve may be adjusted to change its

operation to alter a corresponding shock assembly damping characteristic from a “soft” setting to a “firm” setting (or a stiffness setting somewhere therebetween) by, for example, manually rotating a knob, pushing or pulling a lever, physically manipulating an air pressure control feature, manually operating a cable assembly, physically engaging a hydraulic unit, and the like. For purposes of the present discussion, such instances of manual adjustment/physical manipulation of the valve or component can occur before, during, and/or after “typical operation of the vehicle”.

[0038] It should further be understood that a vehicle suspension may also be referred to using one or more of the terms “passive”, “active”, “semi-active” or “adaptive”. As is typically used in the suspension art, the term “active suspension” refers to a vehicle suspension which controls the vertical movement of the wheels relative to vehicle. Moreover, “active suspensions” are conventionally defined as either a “pure active suspension” or a “semi-active suspension” (a “semi-active suspension” is also sometimes referred to as an “adaptive suspension”). In a conventional “pure active suspension”, a motive source such as, for example, an actuator, is used to move (e.g. raise or lower) a wheel with respect to the vehicle. In a “semi-active suspension”, no motive force/actuator is employed to adjust move (e.g. raise or lower) a wheel with respect to the vehicle.

[0039] Rather, in a “semi-active suspension”, the characteristics of the suspension (e.g. the firmness of the suspension) are altered during typical use to accommodate conditions of the terrain and/or the vehicle. Additionally, the term “passive suspension”, refers to a vehicle suspension in which the characteristics of the suspension are not changeable during typical use, and no motive force/actuator is employed to adjust move (e.g. raise or lower) a wheel with respect to the vehicle. As such, it will be understood that an “active valve”, as defined above, is well suited for use in a “pure active suspension” or a “semi-active suspension”.

[0040] Referring now to FIG. 1A, a perspective view of a bicycle **50** having an active suspension system and an adjustable shock assembly **38** is shown in accordance with an embodiment.

Although a bicycle is used in the discussion. In one embodiment, the adjustable shock assembly **38** could be used on another vehicle such as, but not limited to a road bike, a mountain bike, a gravel bike, an electric bike (e-bike), a hybrid bike, a scooter, a motorcycle, an ATV, a personal water craft (PWC), a four-wheeled vehicle, a snow mobile, a UTV such as a side-by-side, and the like. In one embodiment, the adjustable shock assembly **38** could be used on a suspension inclusive device such as, but not limited to an exoskeleton, a seat frame, a prosthetic, a suspended floor, and the like.

[0041] Thus, between the disclosed examples as provided in view of a bicycle **50**, the disclosed embodiments can be used on adjustable shock assemblies used by vehicles with wheels, skis, tracks, hulls, and/or with suspension inclusive devices such as prosthetic limbs, exoskeletons, and the like.

[0042] In one embodiment, bicycle **50** has a main frame **24** with a suspension system comprising a swing arm **26** that, in use, is able to move relative to the rest of main frame **24**; this movement is permitted by, inter alia, rear adjustable shock assembly **38**. The front fork assembly **34** also provide a suspension function via a front shock assembly **37** in at least one fork leg; as such the bicycle **50** is a full suspension bicycle (such as an all-terrain bike (ATB), mountain bike, e-bike, etc.).

[0043] However, the embodiments described herein are not limited to use on full suspension bicycles. In particular, the term “suspension system” is intended to include vehicles having front suspension only, rear suspension only, seat suspension only, a combination of two or more different suspensions, and the like.

[0044] In one embodiment, swing arm **26** is pivotally attached to the main frame **24** at pivot point **12** which is located above the bottom bracket axis **11**. Although pivot point **12** is shown in a specific location, it should be appreciated that pivot point **12** can be found at different distances from bottom bracket axis **11** depending upon the rear suspension configuration. The use of the specific pivot point **12** herein is provided merely for purposes of clarity. For example, in a hardtail bicycle embodiment, there would be no pivot point **12**. In one embodiment of a hardtail bicycle,

main frame **24** and swing arm **26** would be formed as a fixed frame. Bottom bracket axis **11** is the center of the pedal and front sprocket assembly **13**. Bicycle **50** includes a front wheel **28**, a rear wheel **30** and a seat **32**. A seat **32** is connected to the main frame **24** via a seatpost **33** in order to support a rider of the bicycle **50**.

[0045] Front wheel **28** which is coupled with the front fork assembly **34** via axle **14**. The front fork **34** includes a crown and at least one fork leg. Above the crown, a steerer tube **60** passes through a portion of bicycle main frame **24** and attaches the fork **34** to the handlebars **36** (via a stem) allowing the rider to steer the bicycle **50**. In one embodiment, at least one active valve damper **37** is integrated with fork **34**.

[0046] The rear wheel **30** is connected to the swing arm **26** of the frame **22** at rear axle **15**. A rear damping assembly (e.g., active valve damper **38**) is positioned between the swing arm **26** and the frame **22** to provide resistance to the pivoting motion of the swing arm **26** about pivot point **12**. Thus, the illustrated bicycle **50** includes a suspension member between swing arm **26** and the main frame **24** which operate to substantially reduce rear wheel **30** impact forces from being transmitted to the seat **32** (and thus the rider) of the bicycle **50**.

[0047] In one embodiment, saddle **32** is connected to the main frame **24** via seatpost **33**. In one embodiment, seatpost **33** is a dropper seatpost. In one embodiment, front shock assembly **37**, rear adjustable shock assembly **38**, seatpost **33**, handlebar assembly **36**, and/or the like include one or more active and/or semi-active damping components which are used to reduce an initial force generated by an event (e.g., imparted to a wheel of the vehicle from the surface on (or through) which the vehicle is traveling) to a lesser force as it is transferred to the rest of the vehicle and/or persons riding therein/thereon.

[0048] In one embodiment, bicycle **50** is driven by a chain **19** coupled with both front sprocket assembly **13** and rear sprocket **18**. As the front sprocket assembly **13** is rotated about bottom bracket axis **11**, a force is applied to chain **19** which transfers the energy from the front sprocket assembly **13** to rear sprocket **18**. Chain tension device **17** provides a variable amount of tension on chain **19**. The need for chain **19** length variation can be due to factors such as a number of different gears that may be on one or both of front sprocket assembly **13** and/or rear sprocket **18** and/or changes in chain stay length as the distance between bottom bracket axis **11** (where front sprocket assembly **13** attaches to main frame **24**) and the rear axle **15** changes due to suspension articulation.

[0049] In one embodiment, bicycle **50** includes an active suspension system **75** (shown in FIG. **1B**) consisting of a suspension controller **39**, one or more sensors (e.g., orientationally flexible bump sensor **35f**, **35r**, and the like), smart components, mobile device **95**, active valve dampers (e.g., adjustable shock assembly **38**, front shock assembly **37**, a seat post damper, etc.), power source(s), and the like. In one embodiment, sensor **35r** is positioned on the swing arm **26** the rear axle **15** of bicycle **50**. In one embodiment, sensor **35f** is positioned in an unsprung location of front fork **34**. Additional information for vehicle suspension systems, sensors, and their components as well as adjustment, modification, and/or replacement aspects including manually, semi-actively, semi-actively, and/or actively controlled aspects and wired or wireless control thereof is provided in U.S. Pat. Nos. 8,838,335; 9,353,818; 9,682,604; 9,797,467; 10,036,443; 10,415,662; the content of which are incorporated by reference herein, in their entirety.

[0050] In one embodiment, mobile device **95** is mounted to handlebar assembly **36** of bicycle **50**. Although mobile device **95** is shown mounted to handlebar assembly **36**, it should be appreciated that the mobile device **95** could be in a rider's backpack, pocket, or the like. In general, mobile device **95** is a smart device such as a mobile phone, tablet, a smart phone, a tablet, a smart watch, a piece of smart jewelry, smart glasses, or other user portable device(s) having wireless connectivity. Mobile device **95** is capable of broadcasting and receiving via at least one network, such as, but not limited to, WiFi, Cellular, Bluetooth, NFC, and the like. In one embodiment, mobile device **95** includes one or more of a display, a processor, memory, a location or position system, (such as a global positioning system (GPS), local positioning system, or the like), camera, and one or more

sensors such as audio, visual, motion, acceleration, altitude, and the like.

[0051] In one embodiment, switch **93** is mounted to handlebar assembly **36** of bicycle **50**. In one embodiment, switch **93** is a positional switch used in conjunction with the active suspension system **75**. In one embodiment, switch **93** is a multi-positional switch, an upshift/downshift type of switch, a button type switch, or the like. For example, switch **93** would be a 2-position switch, a 3-position switch, a switch that can cycle through a number of different modes (similar to a gear shift), or the like. Although switch **93** is shown mounted to handlebar assembly **36**, it should be appreciated that switch **93** could be mounted in a different location on the vehicle, on a mount coupled to the vehicle, or the like. In one embodiment, the location of switch **93** is modifiable and is located on the vehicle based on a user's preference.

[0052] In one embodiment, one or a plurality of component(s) of the bicycle **50** are also smart component(s). Examples of the smart component(s) can include one or more of the forks, wheels, rear shocks, front shocks, handlebars, seat posts, pedals, cranks, and the like. In one embodiment, the smart component(s) will include connective features that allow them to communicate wired or wirelessly with one or more of suspension controller **39**, mobile device **95**, one or more sensors **35**, and/or any other smart component(s) within transmission range (thereby becoming connected components).

[0053] In one embodiment, data (including real-time data) is collected or provided from the smart component to the suspension controller **39**. Depending upon the connected component, data such as telemetry attributes to provide angle, orientation, velocity, acceleration, RPM, operating temperature, and the like, can be obtained.

[0054] Referring now to FIG. **1B**, a schematic side view of a bicycle **50** with focus on the active suspension system **75** and adjustable shock assembly **38** (and/or **37**) is shown in accordance with an embodiment.

[0055] In one embodiment, active suspension system **75** includes suspension controller **39**, one or more sensors (e.g., sensor **35f**, **35r**, and the like), hereinafter “sensor **35**”, smart components, mobile device **95**, active valve dampers (e.g., adjustable shock assembly **38**, front shock assembly **37**, a seat post damper, etc.), or the like.

[0056] In general, sensor **35** could be a single sensor (such as an accelerometer) or a combination of sensor types. Sensor **35** is used for sensing characteristics (or changes to characteristics) such as terrain, environment, temperature, vehicle speed, vehicle pitch, vehicle roll, vehicle yaw, component activity, or the like. It is understood that the one or more sensors may be imbedded, moved, mounted, or the like, in any suitable configuration and allowing for any suitable range of adjustment as may be desirable.

[0057] In one embodiment, sensor **35** is a force or acceleration transducer (e.g., strain gage, Wheatstone bridge, accelerometer, hydraulic, interferometer based, optical, thermal or any suitable combination thereof). Further, the sensor **35** may utilize solid state electronics, electro-mechanical principles or MEMS, or any other suitable mechanisms.

[0058] In one embodiment, one, some, or all of the components discussed herein including switch **93**, sensor **35f**, sensor **35r**, suspension controller **39**, active valves, mobile device **95**, and the like are wired and/or wireless and can communicate with one or more of the other components of active suspension system over the wired network, over a wireless network, or over a hybrid wired and wireless network. The wireless network is selected from one or more of: a wireless personal area network (WPAN), a low power network (LPAN), Internet of things (IoT) connectivity, or the like. In one embodiment, the wireless communication protocol could be, but is not limited to, Bluetooth, WiFi, Bluetooth Low Energy (BLE), near field communication (NFC), UHF radio signal, Worldwide Interoperability for Microwave Access (WiMax), long-term evolution (LTE), industrial, scientific, and medical (ISM) band, IEEE 802.15.4 standard communicators, Zigbee, ANT, ISA100.11a (wireless systems for industrial automation: process control and related applications) wireless highway addressable remote transducer protocol (HART), MiWi, IPv6 over low-power

wireless personal area networks (6LoWPAN), thread network protocol, subnetwork access protocol (SNAP), and the like.

[0059] In one embodiment, one, some, or all of the components discussed herein including switch **93**, sensor **35f**, sensor **35r**, suspension controller **39**, active valves, mobile device **95**, and the like, could form a wireless mesh, such as a bicycle area network (BAN) or the like. In one embodiment, one or more components of the BAN could interact with the user/rider in any number of ways such as via touch, sound, vision, radio, wearable, and the like.

[0060] In one embodiment, the components within the wireless mesh may include an auxiliary or propriety private network encryption. In one embodiment, one or more components within the wireless mesh may include communication protocols for one or more peers, such as an out-of-BAN wireless device that doesn't want to share its network. In this case, the out-of-BAN wireless device can provide a hardware interface and it can be piped into the BAN. Thus, in one embodiment, the wireless mesh network can be used to connect and/or control almost any wireless aspect, as the network, topology, and features thereof are well suited to interacting with basic device operating structures.

[0061] In one embodiment, information broadcast from a given component will include a unique identifier (ID) that identifies the specific component that made the broadcast. Thus, even when a number of different components are operating in the same environment, the suspension controller **39** will be able to identify which component sent the signal based on the unique ID. In one embodiment, the unique ID is used during the programming/pairing of the components with suspension controller **39** and/or the network.

[0062] In one embodiment, the unique ID is used by suspension controller **39** to validate the sending component. Although a unique ID is used in one embodiment, in another embodiment, a different identification methodology may be used to identify the different components in the network. In one embodiment, the wireless network is an intra-vehicle wireless network (such as a BAN) for data transmission between at least two components coupled with the vehicle, the at least two components including, but not limited to, at least one sensor, the suspension controller, and at least one peripheral device (such as a smart component, switch, or the like) coupled with the vehicle. In one embodiment, the intra-vehicle wireless network is a wireless mesh network. In one embodiment, the intra-vehicle wireless network includes an intra-vehicle transmission authentication and encryption protocol.

[0063] In one embodiment, the broadcast information or data (e.g., message payload) will include additional information/data comprising the wireless network which is passed to and from peripheral devices in the network. Thus, in one embodiment, the wireless network communication and/or wireless mesh network will allow for information/data to be exchanged between adjacent vehicles, vehicle networks, etc.

[0064] In one embodiment, the wireless network includes an inter-vehicle communication (IVC) wireless network for data transmission between the vehicle and at least another vehicle, between the vehicle and a mobile communications device distinct from the vehicle, between the vehicle and an infrastructure component (such as a traffic light, beacon on a stop sign, road mile marker, a benchmark, or the like). In one embodiment, the IVC wireless network is a wireless mesh network. In one embodiment, the IVC wireless network includes an IVC transmission authentication and encryption protocol.

[0065] The IVC transmission authentication and encryption protocol can be distinct and different from the intra-vehicle transmission authentication and encryption protocol, such that a device receiving a communication can determine the origin of the communication. Often, the origin of the communication is important depending upon the data provided in the communication. For example, a transmission that includes sensor provided information might only be verified and acted upon if it includes the intra-vehicle transmission authentication and encryption protocol (such as for security purposes or the like).

[0066] In one embodiment, the IVC transmission authentication and encryption protocol can include levels of trust. For example, a vehicle used by a friend may have a “trusted” IVC transmission authentication and encryption protocol that allows a sensor from the friend's vehicle to provide sensor data to the user's suspension controller that is verified and acted upon as sensor data from a “trusted” peripheral. In contrast, when an IVC transmission includes sensor provided information but it does not have a “trusted” IVC transmission authentication and encryption protocol it would not be verified and acted upon. However, other information such as stop sign warnings, terrain changing information, or the like from IVC transmissions would be evaluated by the suspension controller and may be used depending upon context, or the like.

[0067] In one embodiment, the sensors **35** of active suspension system **75** provides the obtained sensor data to a suspension controller **39** which uses the sensor data to make suspension adjustments. In one embodiment, suspension controller **39** makes suspension adjustments to adjustable shock assembly **38**, front shock assembly **37**, or the like.

[0068] In one embodiment, location data from a position system is used in conjunction with terrain data to determine probability of obstacles (e.g., to provide a priori knowledge) which is used to optimize the suspension configuration. For example, GPS location information and a terrain database (or the like) is used to establish the bicycle's current location and the terrain about the location (e.g., on a bumpy patch of trail, etc.).

[0069] In one embodiment, suspension controller **39** monitors the sensor(s) **35** for sensor input (and/or the location and terrain data, etc.) and make suspension adjustments in a matter of milliseconds after receiving the sensor data. For example, in one embodiment, sensors on the fork, rear axle, and/or main frame read bump input at the wheel and send the obtained sensor data to the suspension controller **39**. Thus, by placing sensors on the frame and/or in unsprung location for both wheels, the suspension controller **39** processes data from the terrain to constantly adjust the suspension for maximum efficiency and control.

[0070] In one embodiment, the power source for one, some, or all of the suspension controller **39**, sensors **35**, active valves in the adjustable shock assembly **38** and front shock assembly **37**, and the like can be a replaceable battery, a rechargeable battery, a combination thereof, and the like. In one embodiment, the power source is a lithium-ion battery. In one embodiment, the power source is a CR2032 battery. In one embodiment, the power source is a different type of non-rechargeable battery.

[0071] In one embodiment, the power source is a rechargeable battery. In one embodiment, the power source can be recharged wired or wirelessly. For example, a power source having a wirelessly rechargeable capability means it could be charged using a wireless power transfer system. E.g., using an inductive charger (or the like) within a given distance of the wirelessly rechargeable capability of the battery. Examples of a wireless power transfer systems that could be used in one or more embodiments include those defined by the wireless power consortium (WPC) Qi standard, the AirFuel Alliance (e.g., Duracell Powermat, PowerKiss, etc.), WiTricity, and the like.

[0072] In one embodiment, the power source is an energy harvesting switch that does not require a battery or other powered connection. As such, the energy harvesting switch is capable of operating for an indefinite amount of time without requiring any type of recharge, battery change, etc. For example, in one embodiment, the energy harvesting switch utilizes a momentary generator such as ZF electronics AFIG-0007 to provide power.

[0073] Although a number of sensors are shown in FIGS. **1A** and **1B**, it should be appreciated that there may be only a single sensor or two or more sensors in operation. Moreover, in one embodiment, mobile device **95** is part of the active suspension system **75**.

[0074] In general, the one or more sensors may be attached to the swing arm **26** directly, to any link thereof, to an intermediate mounting member, to front fork **34**, to adjustable shock assembly **38**, seat **32**, handlebar assembly **36**, or to any other portion or portions of the bicycle **50** as may be

useful, available, or the like. In one embodiment, one or more sensors and valve actuators (e.g. electric solenoid or linear motor type-note that a rotary motor may also be used with a rotary actuated valve), suspension components, suspension component controller(s) and/or data processing system(s), and the like may be coupled to and/or integrated with the vehicle structure, such as disclosed in U.S. Pat. Nos. 7,484,603; 8,838,335; 8,955,653; 9,303,712; 10,036,443; 10,060,499; 10,443,671; and 10,737,546; the content of which is incorporated by reference herein, in its entirety. Further, sensors and valves, or principles, of patents and other documents incorporated herein by reference, may be integrated one or more embodiments hereof, individually or in combination.

Adjustable Shock Assembly

[0075] With reference now to FIG. 2, a perspective view of adjustable shock assembly **38** is shown in accordance with one embodiment. In one embodiment, adjustable shock assembly **38** includes a top cap portion **100**, shaft end eyelet **105**, lower eyelet **110**, damper body **120**, air sleeve **123**, and reservoir **125**. In FIG. 2, reservoir **125** is shown as an external or piggyback type of reservoir. However, in another embodiment, reservoir **125** may be an internal reservoir. The use of an external reservoir is provided merely for purposes of clarity.

[0076] In one embodiment, adjustable shock assembly **38** has a controller **39** and power source **65** coupled therewith. In one embodiment, the power source **65** is integrated with controller **39**. In one embodiment, power source **65** is separate from controller **39**. In one embodiment, by providing controller **39** and power source **65** on the shock assembly **38**, shock assembly **38** becomes a self-contained active valve adjustable shock assembly. In one embodiment, adjustable shock assembly **38** may include one or more sensors such that information from the sensors can be provided to controller **39** providing a fully self-contained active adjustable shock assembly **38**.

[0077] In one embodiment, controller **39** is similar to the controller **39** described herein. In one embodiment, power source **65** is similar to the power sources described herein. For example, it may be a replaceable battery, a rechargeable battery, an e-bike battery, or combination thereof, and the like. In one embodiment, power source **65** is a lithium-ion battery. In one embodiment, power source **65** is a CR2032 battery. In one embodiment, the power source is a different type of non-rechargeable battery.

[0078] In one embodiment, power source **65** is a rechargeable battery which can be recharged wired or wirelessly. In one embodiment, power source **65** is an energy harvesting switch that does not require a battery or other powered connection. For example, in one embodiment, power source **65** is a momentary generator such as ZF electronics AFIG-0007.

[0079] In one embodiment, adjustable shock assembly **38** includes one or more adjustable features such as, but not limited to, a pneumatic valve **230** (e.g., a Shrader valve, Presta valve, Dunlop valve, or the like), an adjustable active valve assembly **205**, a compression adjuster **210**, and an adjustable rebound valve assembly **215**. In one embodiment, adjustable shock assembly **38** includes each of the pneumatic valve **230**, the adjustable active valve assembly **205**, the compression adjuster **210**, and the adjustable rebound valve assembly **215**.

[0080] In one embodiment, adjustable shock assembly **38** may include fewer components, e.g., adjustable shock assembly **38** may not include most, some, or one of the top cap portion **100**, shaft end eyelet **105**, damper body **120**, reservoir **125**, pneumatic valve **230**, adjustable active valve assembly **205**, compression adjuster **210**, and adjustable rebound valve assembly **215**.

[0081] In its basic form, adjustable shock assembly **38** controls the speed of movement of the piston shaft **130** (of FIG. 3) by metering incompressible fluid from one side of the main piston **131** (of FIG. 3) to the other, and additionally from the main chamber **121** (of FIG. 3) to the reservoir **125**, during a compression stroke (and in reverse during the rebound or extension stroke) to account for the added piston shaft **130** volume.

[0082] In one embodiment, the adjustable shock assembly **38** is a stand-alone fluid damper assembly, a coil sprung adjustable shock assembly, an air sprung fluid damper assembly, or the

like.

[0083] With reference now to FIG. 3, a side view of the upper portion of adjustable shock assembly **38** including a cutaway view of the damper body **120** and the adjustable rebound valve assembly **215** is shown in accordance with one embodiment. For purposes of clarity, components of adjustable shock assembly **38** shown in FIG. 3 which are similar to those already discussed in FIG. 2 are not repeated. However, the discussion of FIG. 2 is incorporated by reference. In one embodiment, aspects, components, and views that are shown in more detail in FIG. 3 or include additional operational details will be discussed.

[0084] In one embodiment, the cutaway view of the damper body **120** shows main piston **131**, piston shaft **130**, bearing housing **172**, main chamber **121** of damper body **120**, and fluid flow path **340** from main chamber **121**, through compression adjuster **210**, through adjustable active valve assembly **205**, and to reservoir **125**. In one embodiment, fluid flow path **340** travels through the inside of a portion of piston shaft **130**. In one embodiment, fluid flow path **340** does not travel through the inside of a portion of piston shaft **130**.

[0085] In one embodiment, the cutaway view of the adjustable rebound valve assembly **215** shows the rebound needle adjustable feature **324** (e.g., a manual knob, switch, lever, etc. or an electronic component such as a solenoid, stepper motor, electric motor, or the like) near shaft-end eyelet **105**. In one embodiment, the rebound adjustment needle **323**, which is adjusted by the rebound needle adjustable feature **324** is located within the hollow piston shaft **130**.

[0086] During a compression stroke, main piston **131** and shaft **130** will move into the main chamber **121**. Within main chamber **121** is a working fluid (or fluid). While the fluid will be able to pass through shims in main piston **131** and/or use a bypass to go around main piston **131** from the compression side to the rebound side within main chamber **121**, the volume of the shaft **130** that moves into the main chamber **121** will need to displace the incompressible fluid. Therefore, as the main piston and shaft move into the main chamber, the building pressure will force shaft-displaced fluid to travel along one or more fluid flow paths such as fluid flow path **340**, through compression adjuster **210**, through adjustable active valve assembly **205** and into reservoir **125**.

[0087] In contrast, during a rebound stroke, main piston **131** and shaft **130** will move out of the main chamber **121**. Here again, while the fluid will be able to pass through shims in main piston **131** and/or use a bypass to go around main piston **131** this time from the rebound side to the compression side within main chamber **121**, the volume of the shaft **130** that moves out of main chamber **121** will cause a reduced pressure. Therefore, as the main piston and shaft move out of the main chamber, the reduced pressure will draw fluid from reservoir **125**, through adjustable rebound valve assembly **215**, and through fluid flow path **340** back into main chamber **121**.

[0088] With reference now to FIG. 4A, a side view of the upper portion of adjustable shock assembly **38** including a cutaway view of the adjustable active valve assembly **205** and a cutaway view of the compression adjuster **210** is shown in accordance with one embodiment. For purposes of clarity, components of adjustable shock assembly **38** shown in FIG. 4A which are similar to those already discussed in FIGS. 2 and 3 are not repeated. However, the discussion of FIGS. 2 and 3 are incorporated by reference. In one embodiment, aspects, components, and views that are shown in more detail in FIG. 4A or include additional operational details will be discussed.

[0089] In one embodiment, compression adjuster **210** includes adjustable valve **511**, compression valve stack **510**, and knob **512** for manually adjusting adjustable valve **511**. In one embodiment, adjustable valve **511** is a low-speed compression (LSC) adjustable valve. In one embodiment, compression valve stack **510** is a high-speed compression valve stack. In general, low-speed adjustments are used to adjust body motion, shock shaft velocity such as zero to twenty or thirty inches per second. In contrast, high-speed adjustments are used to deal with shock shaft velocities that are greater than twenty to thirty inches per second, such as square edge impacts, and the like. Thus, low-speed adjustments are usually (but not only) used to adjust body motion, while high-speed adjustments are usually (but not only) used to adjust impacts. In one embodiment, an adjuster

may adjust across a different range that could include some (or all) of both low-speed and high-speed adjustment ranges.

[0090] In one embodiment, adjustable valve **511** may be adjusted by a powered component such as a solenoid, stepper motor, electric motor, or the like. In one embodiment, adjustable valve **511** may be adjusted both manually such as by knob **512** and electrically such as by a solenoid, stepper motor, electric motor, or the like. In one embodiment, adjustable valve **511** (or knob **512**) includes one or more hard stops to ensure the adjustable valve **511** isn't rotated too far in either direction. For example, the stops will ensure a user (or electric motor) doesn't over-rotate and deleteriously affect the operation of adjustable valve **511**.

[0091] In one embodiment, knob **512** will include numbers, arrows, and/or other identifying features that will allow a user to visually identify one or more predefined settings, directions to turn the knob for different performance aspects, and the like. In one embodiment, knob **512** will include dents, grooves, or the like that will provide haptic feedback to a user turning the knob. In one embodiment, knob **512** will include a range shifting capability to allow the range of the adjustment of adjustable valve **511** to be modified.

[0092] In one embodiment, the shaft-displaced fluid flow path shown in FIG. **3** enters into compression adjuster **210** as indicated by flow path arrows **341**. At that point, the flow of shaft-displaced fluid will follow one or both of parallel flow paths **342** and **343**. In one embodiment, parallel flow path **342** includes adjustable valve **511** while parallel flow path **343** includes compression valve stack **510**. The operation of compression adjuster **210** and its associated fluid flow are described in more detail in the schematic discussion herein.

[0093] In one embodiment, adjustable active valve assembly **205** includes active valve **521**, firm mode blowoff stack **525**, firm mode adjuster **526**, and knob **527** for manually adjusting firm mode adjuster **526**. In one embodiment, firm mode adjuster **526** may be adjusted by a powered component such as a solenoid, stepper motor, electric motor, or the like. In one embodiment, firm mode adjuster **526** may be adjusted both manually such as by knob **527** and electrically such as by a solenoid, stepper motor, electric motor, or the like. In one embodiment, firm mode adjuster **526** (or knob **527**) includes one or more stops to stop the firm mode adjuster **526** from rotating too far in either direction. For example, the stops will ensure a user (or electric motor) doesn't over-rotate and deleteriously affect the operation of firm mode adjuster **526**, such as by setting the blowoff pressure of firm mode blowoff stack **525** past its operational limits, and the like.

[0094] In one embodiment, knob **527** will include numbers, arrows, and/or other identifying features that will allow a user to visually identify one or more predefined settings, directions to turn the knob for different performance aspects, and the like. In one embodiment, knob **527** will include dents, grooves, or the like that will provide haptic feedback to a user turning the knob. In one embodiment, knob **527** will include a range shifting capability to allow the range of the adjustment of firm mode adjuster **526** to be modified.

[0095] In one embodiment, the shaft-displaced fluid flow enters into adjustable active valve assembly **205** from compression adjuster **210** as indicated by flow path arrows **344**. At that point, the flow of shaft-displaced fluid will follow one or both of parallel flow paths **346** and **347**. In one embodiment, parallel flow path **346** includes active valve **521** while parallel flow path **347** includes firm mode blowoff stack **525**. The operation of adjustable active valve assembly **205** and its associated fluid flow are described in more detail in the schematic discussion herein.

[0096] In one embodiment, adjustable rebound valve assembly **215** also includes a spring loaded shim **217** and knob **216**. In one embodiment, spring loaded shim **217** is used to keep the rebound flow path closed during the compression stroke. However, when the shock assembly **38** starts a rebound stroke, the fluid will flow back from reservoir **125** lift the spring loaded shim **217** and flow along flow path **340** into main chamber **121**.

[0097] In one embodiment, adjustable rebound valve assembly **215** may be adjusted both manually such as by knob **216** and electrically such as by a solenoid, stepper motor, electric motor, or the

like. In one embodiment, knob **216** includes one or more stops to stop a user from rotating adjustable rebound valve assembly **215** too far in either direction. For example, the stops will ensure a user (or electric motor) doesn't over-rotate knob **216** deleteriously affecting the operation of adjustable rebound valve assembly **215**, such as by setting the adjustable rebound valve assembly **215** past its operational limits, or the like.

[0098] In one embodiment, knob **216** will include numbers, arrows, and/or other identifying features that will allow a user to visually identify one or more predefined settings, directions to turn the knob for different performance aspects, and the like. In one embodiment, knob **216** will include dents, grooves, or the like that will provide haptic feedback to a user turning the knob. In one embodiment, knob **216** will include a range shifting capability to allow the range of the adjustment of adjustable rebound valve assembly **215** to be modified.

[0099] With reference now to FIG. **4B**, a side view of adjustable shock assembly **38** including a cutaway portion of reservoir **125** and a cutaway portion of adjustable active valve assembly **205** is shown in accordance with one embodiment. For purposes of clarity, components of adjustable shock assembly **38** shown in FIG. **4B** which are similar to those already discussed in FIGS. **2**, **3** and **4A** are not repeated. However, the discussion of FIGS. **2**, **3**, and **4A** are incorporated by reference. In one embodiment, aspects, components, and views that are shown in more detail in FIG. **4B** or include additional operational details will be discussed.

[0100] In one embodiment, the shaft-displaced fluid flow enters into reservoir **125** from adjustable active valve assembly **205** as indicated by flow path arrow **351**.

[0101] In one embodiment, reservoir **125** includes an internal floating piston (IFP) **126** that separates the working fluid in the reservoir **125** from an amount of pressurized gas. In general, the reservoir is used to store any shaft-displaced fluid due to a compression event that causes the piston shaft **130** (of FIG. **3**) to move into the main chamber **121** of damper body **120**. As described herein, when the piston shaft **130** moves into main chamber **121** it reduces the volume available in main chamber **121**. Prior to the piston shaft **130** incursion, that volume was used to contain the working fluid, during the piston shaft **130** incursion, the reduced volume results in the displacement of the fluid from the main chamber **121** to the reservoir **125** (i.e., shaft-displaced fluid). In one embodiment, fluid communication between the main chamber **121** of damper body **120** and the reservoir **125** may be via fluid flow pathways controlled by one, two, or all three of the adjustable active valve assembly **205**, compression adjuster **210** and adjustable rebound valve assembly **215**. [0102] Additional descriptions and details of reservoir **125**, including at least a reservoir housing and IFP **126**, are described in U.S. Pat. No. 7,374,028 the content of which is incorporated by reference herein, in its entirety.

Schematic Diagram

[0103] With reference now to FIG. **5**, a schematic diagram **500** of the fluid flow in the adjustable shock assembly **38** is shown in accordance with an embodiment. For purposes of clarity, the schematic diagram **500** of FIG. **5** is discussed in conjunction with the features of FIGS. **2-4B**. In one embodiment, schematic diagram **500** includes main chamber **121**, compression adjuster **210**, adjustable active valve assembly **205**, reservoir **125**, and adjustable rebound valve assembly **215**.

[0104] In one embodiment, when shock assembly **38** is undergoing a compression stroke, the shaft-displaced fluid will flow from the main chamber **121**, and into compression adjuster **210**.

[0105] In one embodiment, compression adjuster **210** includes at least two parallel flow paths and provides an adjustable damping capability when shock assembly **38** is in an open mode. In one embodiment, one of the parallel flow paths includes an adjustable valve **511** flow control feature while the other parallel flow path includes a compression valve stack **510** flow control feature.

[0106] Once the fluid has passed through one or more of the at least two parallel flow paths, the fluid will leave compression adjuster **210**. In one embodiment, the two parallel fluid flow pathways join back together as (or before) the fluid leaves compression adjuster **210**.

[0107] Once the fluid leaves compression adjuster **210** it continues traversing the flow path(s) until

some or all of it enters adjustable active valve assembly **205**.

[0108] In one embodiment, adjustable active valve assembly **205** includes at least two parallel flow paths. In one embodiment, one of the parallel flow paths includes an active valve **521** flow control feature while the other parallel flow path includes both a firm mode blowoff stack **525** and a firm mode adjuster **526** flow control features.

[0109] Once the fluid has passed through one or more of the at least two parallel flow paths, the fluid will leave adjustable active valve assembly **205**. In one embodiment, the two parallel fluid flow pathways join back together as (or before) the fluid leaves adjustable active valve assembly **205**.

[0110] After the fluid leaves adjustable active valve assembly **205** it will continue along one or more fluid pathways until it reaches reservoir **125**. In one embodiment, reservoir **125** includes IFP **126** separating the fluid from a gas charge.

[0111] During a rebound stroke, the fluid flows from the reservoir **125** along one or more rebound fluid pathways back to main chamber **121**. In one embodiment, the rebound fluid pathway(s) are controlled by an adjustable rebound valve assembly **215**. In one embodiment, adjustable rebound valve assembly **215** includes a compression check valve **217** such that the fluid can bypass the compression valve assemblies (e.g., adjustable active valve assembly **205** and compression adjuster **210**) when flowing from the reservoir **125** to the main chamber **121** during the rebound stroke. In one embodiment adjustable rebound valve assembly **215** is a radially varying compression adjuster. In one embodiment, adjustable rebound valve assembly **215** may be similar to adjustable active valve assembly **205** or compression adjuster **210** thereby providing adjustable rebound damping characteristics.

[0112] Although schematic diagram **500** shows different configurations for and/or components for compression adjuster **210**, adjustable active valve assembly **205**, and adjustable rebound valve assembly **215**, the schematic is used to represent one or more embodiments and is provided for purposes of clarity. However, it should be appreciated that in other embodiment, different flow control features may be used in the different pathways.

[0113] For example, with respect to compression adjuster **210**, in one embodiment, compression adjuster **210** includes at least two parallel flow paths. In one embodiment, one of the parallel flow paths will include the adjustable valve **511** (or the active valve **521**) flow control feature while the other parallel flow path will include both the compression valve stack **510** flow control feature and a compression valve stack adjuster (similar in functionality/operation to the firm mode adjuster **526** such that the firmness of the compression valve stack **510** may be adjusted manually and/or automatically) flow control features.

[0114] In yet another embodiment, compression adjuster **210** will include at least three parallel flow paths. In one embodiment, one of the parallel flow paths will include the adjustable valve **511** (or the active valve **521**) flow control feature, another of the parallel flow paths will include the compression valve stack **510** flow control feature, and yet another parallel flow path will include both the compression valve stack **510** and the compression valve stack adjuster flow control features.

[0115] In yet another embodiment, compression adjuster **210** will include at least four parallel flow paths. In one embodiment, one of the parallel flow paths will include the adjustable valve **511** flow control feature, another of the parallel flow paths will include the active valve **521** flow control feature, another of the parallel flow paths will include the compression valve stack **510** flow control feature, and yet another parallel flow path will include both the compression valve stack **510** and the compression valve stack adjuster flow control features.

[0116] Similarly, with respect to adjustable active valve assembly **205**, in one embodiment, adjustable active valve assembly **205** includes at least two parallel flow paths. In one embodiment, one of the parallel flow paths will include the active valve **521** (or the adjustable valve **511**) flow control feature while the other parallel flow path will include the firm mode blowoff stack **525**

without the firm mode adjuster **526** flow control features.

[0117] In another embodiment, adjustable active valve assembly **205** will include at least three parallel flow paths. In one embodiment, one of the parallel flow paths will include the active valve **521** (or the adjustable valve **511**) flow control feature, another of the parallel flow paths will include the firm mode blowoff stack **525** without the firm mode adjuster **526** flow control features, and yet another parallel flow path will include both the firm mode blowoff stack **525** and the firm mode adjuster **526** flow control features.

[0118] In yet another embodiment, adjustable active valve assembly **205** will include at least four parallel flow paths. In one embodiment, one of the parallel flow paths will include the active valve **521** flow control feature, another of the parallel flow paths will include the adjustable valve **511** flow control feature, another of the parallel flow paths will include the firm mode blowoff stack **525** without the firm mode adjuster **526** flow control features, and yet another parallel flow path will include both the firm mode blowoff stack **525** and the firm mode adjuster **526** flow control features.

[0119] With respect to adjustable rebound valve assembly **215**, in different embodiments, adjustable rebound valve assembly **215** may include any of the flow paths and features recited in the embodiments of compression adjuster **210** and/or adjustable active valve assembly **205** disclosed herein.

[0120] It should also be appreciated that in different embodiments, shock assembly **38** may include any variation of the compression adjuster **210** described herein (including the number of parallel pathways and their associated flow control features), as well as any variation of the adjustable active valve assembly **205** described herein (including the number of parallel pathways and their associated flow control features), and any variation of the adjustable rebound valve assembly **215** described herein (including the number of parallel pathways and their associated flow control features).

[0121] For example, the variations to one, some, or all of compression adjuster **210**, adjustable active valve assembly **205**, and/or adjustable rebound valve assembly **215** may be used to provide different levels/adjustability ranges/performance results, and the like for the compression and/or rebound tuning characteristics for similar or different vehicles. The different iterative variations of one, some, or all of compression adjuster **210**, adjustable active valve assembly **205**, and/or adjustable rebound valve assembly **215** would be driven by aspects such as, but not limited to, vehicle cost, vehicle use/performance requirements (e.g., on-road, off-road, gravel/dirt roads/trails/paths, downhill terrain, rough terrain (e.g., rocky, rutty, obstacle inclusive, sand, etc.) water, air, terrestrial, extra-terrestrial, etc.), and the like.

[0122] Thus, as described in further detail below, and as shown in schematic diagram **500** (and FIGS. **2-4B**) the amount of damping and on-the-fly damping adjustability available to shock assembly **38** is no longer dependent upon either adjustable active valve assembly **205**, compression adjuster **210**, or any other single component. Instead, the amount of damping and on-the-fly damping adjustability available to shock assembly **38** is enhanced by the combined, individually tunable characteristics, of adjustable active valve assembly **205** and/or compression adjuster **210**.

[0123] For example, the damping provided by adjustable active valve assembly **205** is modified and/or biased by the additional damping changes provided by the combination of the fluid passing through compression adjuster **210** and then through adjustable active valve assembly **205** (or vice-versa, e.g., the fluid passing through adjustable active valve assembly **205** and then through compression adjuster **210**). For example, in one embodiment, a user is able to bias the closed (or partially closed) mode of one of adjustable active valve assembly **205** or compression adjuster **210** with an open (or partially open) mode of the other of adjustable active valve assembly **205** or compression adjuster **210**. Such a capability is useful in allowing a user (or controller, application, computing device, or the like) to obtain further compression (and/or rebound) control granularity of the damping characteristics of shock assembly **38**.

[0124] In one embodiment, compression adjuster **210** and adjustable active valve assembly **205** may be arranged in parallel as opposed to the depicted serial configuration.

[0125] In one embodiment, active valve **521**, adjustable valve **511**, firm mode adjuster **526**, compression valve stack adjuster flow control, and/or other components of shock assembly **38** are electronically and/or manually adjustable to include being turned on, turned off, opened, partially opened, closed, or the like.

[0126] In one embodiment, one, some, or all of active valve **521**, adjustable valve **511**, firm mode adjuster **526**, compression valve stack adjuster flow control, and/or other components of shock assembly **38** include an externally adjustable feature, such as a knob, button, lever, switch, or the like which can be manually manipulated by a user to change one or a plurality of damping characteristics of shock assembly **38**.

[0127] In one embodiment, one, some, or all of active valve **521**, adjustable valve **511**, firm mode adjuster **526**, compression valve stack adjuster flow control, and/or other components of shock assembly **38** include one or more electronically adjustable features, such as a solenoid, stepper motor, electric motor, or the like to seat or unseat a valve needle (or poppit, plug, etc.) with respect to a valve seat (or orifice).

[0128] In one embodiment, the active valve **521**, adjustable valve **511**, firm mode adjuster **526**, compression valve stack adjuster flow control, and/or other components of shock assembly **38** will receive one or more adjustment. The one or more adjustment inputs will cause the electronically adjustable component to adjust the position of the valve needle with respect to the valve seat (which will cause the fluid pathway flow characteristics to change as the fluid pathway will be closed, opened, partially opened, or partially closed), thereby modifying one or more damping characteristics of shock assembly **38**.

[0129] In one embodiment, the adjustment inputs are received via a wired or wireless electronic communication from one or more devices such as, but not limited to, controller **39**, switch **93** (or the like), mobile device **95** (e.g., a laptop, tablet, computer, smart watch, smart jewelry), sensor **35**, or the like as discussed previously. In one embodiment, the device providing the adjustment input may be integrated with shock assembly **38**.

Adjustable Active Valve Assembly **205**

[0130] With reference now to FIGS. **4A**, **4B**, and **5**, in one embodiment, adjustable active valve assembly **205** includes an active valve **521**, a firm mode blowoff stack **525**, and a firm mode adjuster **526** to adjust the blowoff pressure of firm mode blowoff stack **525**.

[0131] In one embodiment, firm mode adjuster **526** is used to (manually, electronically, remotely, or a combination thereof) adjust the blowoff pressure of firm mode blowoff stack **525**.

[0132] Although a number of electronically adjustable components have been described herein, for purposes of clarity, the following discussion will refer to the electronically adjustable components in an embodiment of a solenoid to seat or unseat a needle with respect to a valve seat. However, it should be appreciated that in other embodiments, other types of electronically adjustable components may be used. The use of the solenoid, needle, and valve seat are provided as one embodiment and for purposes of clarity.

[0133] When the active valve **521** is in a closed configuration, the needle is seated against the valve seat such that the fluid flow through the parallel flow path including active valve **521** would be in its most restricted flow state. In so doing, a substantial volume (e.g., a preponderance, majority, most, all, etc.) of the fluid flowing through adjustable active valve assembly **205** would be flowing through the parallel flow path including firm mode blowoff stack **525**.

[0134] In contrast, when the active valve **521** is in a fully (or nearly fully) open configuration, the needle is moved its furthest distance away from the valve seat and a substantial volume (e.g., most, a substantial volume e.g., most, a preponderance, a majority, or possibly all) of the fluid that flows through adjustable active valve assembly **205** is able to flow through the parallel flow path including fully (or nearly fully) open active valve **521** and, as such, may not require the use of the

parallel flow path including firm mode blowoff stack **525**.

[0135] In one embodiment, active valve **521** only has a closed position and a full-open (or nearly full-open) position.

[0136] In one embodiment, active valve **521** will include a number of positions including the closed position and the full-open (or nearly full-open) position. In general, by adjusting the distance between the needle and the valve seat, the percentage of the fluid flow rate that can flow through the parallel flow path including active valve **521** and thereby bypass the parallel flow path including firm mode blowoff stack **525** is also adjustable.

[0137] In one embodiment, there may be predefined number of positions, e.g., closed, half-open, fully open (or nearly full-open). In such an embodiment, the position of the needle with respect to the valve seat will selectively be: fully seated (closed), halfway or partially unseated (partially open), or fully unseated (full-open or nearly full-open).

[0138] In one embodiment, the active valve **521** is infinitely adjustable. That is, the distance between the needle and the valve seat can be adjusted to any point between the closed (or minimum flow rate) position and the full-open (or nearly full-open) position. In so doing, the percentage of the fluid flow rate that flows through the parallel flow path including active valve **521** and the parallel flow path including firm mode blowoff stack **525** can be infinitely adjusted thereby providing nuanced adjustment capabilities of the damping characteristics of shock assembly **38**. In one embodiment, adjustable active valve assembly **205** (or one or more components thereof) can be adjusted by inputs received from the user by physical manipulation or via electronic transmission and/or automatically by automatic inputs generated by the controller **39**, sensor **35**, and/or other electronic devices.

[0139] In one embodiment, the fixed size of the flow path(s) and openings (e.g., orifice openings) of the active valve **521** provide a high fluid flow rate without causing an excessive pressure drop. In other words, when completely opened, the parallel flow path including active valve **521** is capable of supporting the flow of a substantial volume (e.g., all, most, a preponderance, etc.) of the fluid passing through adjustable active valve assembly **205** such that no pressure drop is built (or occurs), while also providing a complete (or nearly complete) bypass around the parallel flow path including firm mode blowoff stack **525**.

[0140] For example, with reference to schematic diagram **500**, when active valve **521** is fully open (or nearly fully-open), displaced fluid flows from the main chamber **121** through compression adjuster **210** and then into adjustable active valve assembly **205** where the full (or nearly full) flow (or in other embodiments, more than three-quarters of the flow, more than two-thirds of the flow, or the like.) of the fluid is able to pass through the parallel flow path including active valve **521** and continue into the reservoir **125**. Thus, when active valve **521** is in the fully open (or nearly fully-open) state, there is a reduced amount (e.g., little or no) of restriction to the fluid flow as it moves through the adjustable active valve assembly **205**.

[0141] In contrast, when active valve **521** is closed, displaced fluid still flows from the main chamber **121** through compression adjuster **210** and then into adjustable active valve assembly **205**. However, at adjustable active valve assembly **205**, the fluid is blocked from flowing through the parallel flow path including closed active valve **521** and instead has to pass through the parallel flow path including firm mode blowoff stack circuit **525** before continuing into the reservoir **125**.

[0142] The closed (or partially closed) active valve **521** will impede or slow the total amount of fluid flowing through adjustable active valve assembly **205** thereby reducing the amount of shaft-displaced fluid that is exiting the main chamber **121** through fluid pathway **340**. This reduction in the amount of shaft-displaced fluid exiting the main chamber **121** (for a given pressure) will slow the movement of main piston **131** into the main chamber **121**. However, while the movement of main piston **131** is slowed, it is still in compression and still moving. As such, the pressure being applied to the fluid by the increasing volume of the shaft **130** as it continues to move into the main chamber **121** will increase. The increased pressure on the fluid in the main chamber will increase

the pressure that the shaft-displaced fluid is applying to the firm mode blowoff stack circuit **525** of adjustable active valve assembly **205** which will allow more fluid to pass therethrough. In so doing, the compression damping firmness of shock assembly **38** will be increased.

[0143] Similarly, when the active valve **521** is moved from a closed (or partially closed) state to an opened (or partially opened) state, the amount of fluid that can flow through active valve **521** over a given time period (for a given pressure) will increase (as more fluid can bypass the firm mode blowoff stack circuit **525**). This will result in an increase in the amount of shaft-displaced fluid that is able to exit the main chamber **121** through fluid pathway **340** (for a given pressure) which will allow main piston **131** and shaft **130** to more easily move into the main chamber **121** as the fluid pressure caused by the increasing volume of the shaft **130** as it continues to move into the main chamber **121** will have been reduced. In so doing, the compression damping firmness of shock assembly **38** will be decreased.

[0144] In one embodiment, firm mode blowoff stack **525** includes firm-mode adjuster **526** which may have an externally adjustable user manipulatable features (e.g., a knob, switch, lever, etc.) and/or an electrically controlled solenoid that allows a user to manually adjust (and/or a controller, sensor, computer device, or the like to automatically adjust) the blowoff pressure (or other characteristics) of the firm mode blowoff stack **525** and thus the damping characteristics of shock assembly **38**.

[0145] For example, if active valve **521** is full-open (or nearly full-open), there is a reduced amount (similar to the flow volume described in the lockout description herein) shaft-displaced fluid flowing through the parallel flow path including firm mode blowoff stack **525**, and as such, any changes made to the blowoff pressure of firm mode blowoff stack **525** by firm-mode adjuster **526** would be immaterial.

[0146] However, if active valve **521** is not full-open (or nearly full-open), there will be some amount of shaft-displaced fluid flowing through the parallel flow path including firm mode blowoff stack **525**. As such, firm mode adjuster **526** could be adjusted manually and/or electronically via a user or automatically, to increase or decrease the blowoff pressure of the firm mode blowoff stack **525**. In one embodiment, if firm mode adjuster **526** increases the blowoff pressure of the firm mode blowoff stack **525**, then similar to the closed (or partially closed) active valve **521** example above, the shaft-displaced fluid flow will decrease (or the pressure necessary to move the fluid through the firm mode blowoff stack **525** will increase) causing the compression damping characteristics of shock assembly **38** to become firmer.

[0147] In contrast, if firm mode adjuster **526** decreases the blowoff pressure of the firm mode blowoff stack **525**, then similar to the example above where the active valve **521** is moved from a closed (or partially closed) state to partially opened (or more partially opened) state, the shaft-displaced fluid flow will increase (or the pressure necessary to move the fluid through the firm mode blowoff stack **525** will decrease) causing the compression damping characteristics of shock assembly **38** to become softer.

Compression Adjuster **210**

[0148] With reference now to FIGS. **4A** and **5**, in one embodiment, compression adjuster **210** includes an adjustable valve **511** and a compression valve stack **510**. In one embodiment, adjustable valve **511** is manually adjustable from a minimum flow configuration to a high flow configuration without an excessive pressure drop. In one embodiment, adjustable valve **511** is automatically adjustable from a minimum flow configuration to a high flow configuration without an excessive pressure drop. In one embodiment, adjustable valve **511** can include a combination of manually adjustable and automatically adjustable valving to adjust the flow rate therethrough from a minimum flow configuration to a high flow configuration without an excessive pressure drop. Unless otherwise stated, the operation of adjustable valve **511** is similar to that of active valve **521** which is incorporated by reference herein, and as such is not repeated in its entirety for purposes of clarity.

[0149] In one embodiment, when adjustable valve **511** is closed, a substantial volume (e.g., all, almost all, more than three-quarters of the flow, more than two-thirds of the flow, or the like) of the fluid that flows through compression adjuster **210** will have to flow through the parallel flow path including compression valve stack **510**. In contrast, when the adjustable valve **511** is fully open (or nearly fully-open) the fluid moving through compression adjuster **210** will flow through the parallel flow path including adjustable valve **511** and mostly (or completely) bypass the parallel flow path including compression valve stack **510**. When adjustable valve **511** is partially opened some of the fluid will flow through the parallel flow path including adjustable valve **511** and some of the fluid will flow through the parallel flow path including compression valve stack **510**. Thus, similar to the operation of adjustable active valve assembly **205** described above, in one embodiment, the low-speed compression characteristics of shock assembly **38** can be manually and/or automatically adjusted by opening, closing, or partially closing adjustable valve **511**.

[0150] In one embodiment, with reference to schematic diagram **500**, when the adjustable valve **511** is open, displaced fluid flows into compression adjuster **210** where the full (or nearly full) flow (or in other embodiments, more than three-quarters of the flow, more than two-thirds of the flow, or the like) will pass through the parallel flow path including adjustable valve **511** and bypass the parallel flow path including compression valve stack **510**. The fluid will then flow into and through adjustable active valve assembly **205** before it continues into reservoir **125**. Thus, in the open state of adjustable valve **511**, there are reduced restrictions on the fluid flow as it passes through compression adjuster **210**. In the open state, the compression damping firmness of shock assembly **38** will be decreased.

[0151] In contrast, when the adjustable valve **511** is closed, displaced fluid is blocked from flowing through the parallel flow path including adjustable valve **511** and will have to flow through the parallel flow path including compression valve stack **510** before it can leave compression adjuster **210**.

[0152] Similarly, when the adjustable valve **511** is partially opened, some of the displaced fluid will be able to flow through the parallel flow path including adjustable valve **511** and some of the displaced fluid will have to flow through the parallel flow path including compression valve stack **510** before it can leave compression adjuster **210**.

[0153] Thus, by modifying the opening of adjustable valve **511**, a user and/or an electronic device can manually and/or automatically control the amount of fluid that is able to flow unrestrictedly through the parallel flow path including adjustable valve **511** and consequentially the amount of shaft-displaced fluid that must flow through the parallel flow path including compression valve stack **510**. This control of the fluid flow rate (for a given pressure) through the parallel flow paths will allow the user and or the electronic device to increase or decrease the firmness of shock assembly **38** depending upon whether the flow is more or less restricted.

[0154] Therefore, similar to the discussion above with respect to active valve **521**, adjusting the opening of adjustable valve **511** to a larger (e.g., full-open or nearly full-open) size will cause a decrease in the compression damping firmness of shock assembly **38**. In contrast, adjusting the opening of adjustable valve **511** to a smaller (or closed) size will cause an increase in the compression damping firmness of shock assembly **38**.

[0155] As such, compression adjuster **210** will provide adjustable compression damping which can be used to create a softer or firmer characteristic of shock assembly **38** during the compression stroke.

[0156] In one embodiment, adjustable rebound valve assembly **215** includes a compression check valve such that the fluid can only flow from the reservoir **125** to the main chamber **121** during the rebound stroke. In one embodiment adjustable rebound valve assembly **215** is a radially varying compression adjuster. In one embodiment, adjustable rebound valve assembly **215** may be similar to adjustable active valve assembly **205** or compression adjuster **210** thereby providing adjustable rebound damping characteristics.

[0157] In one embodiment, by placing the compression adjuster **210** and the adjustable active valve assembly **205** in series, a user can manually (e.g., externally and/or electronically) adjust a number of damping characteristic of the shock assembly **38** on the fly. In one embodiment, by placing the compression adjuster **210** and the adjustable active valve assembly **205** in series, a controller or other electronic device can automatically adjust a number of damping characteristic of the shock assembly **38** on the fly. In one embodiment, by placing the compression adjuster **210** and the adjustable active valve assembly **205** in series, a user and a controller, or other electronic device, can manually and/or automatically adjust one or both of the open characteristic and closed characteristic of the shock assembly **38**.

[0158] Moreover, in one embodiment, a number of damping characteristics that were previously only adjustable by tearing down and modifying the shock assembly **38** are now available to be adjusted on the fly manually and/or automatically. In addition, since the compression adjuster **210** and the adjustable active valve assembly **205** are in series, even finer tuning of the compression damping characteristics of shock assembly **38** is possible, as each of the compression adjuster **210** and the adjustable active valve assembly **205** are capable of being adjusted independently and then further adjusted based on the added interaction that is now occurring therebetween. In other words, the closed mode damping is additive to the setting selected for the open mode state which can be more intuitive for the user.

Initial Set Up/Tuning

[0159] The following is one example for tuning the shock assembly **38** having the compression adjuster **210** and the adjustable active valve assembly **205** in series. Initially, the user will turn off the controller **39** which will open the active valve **521** and put the adjustable active valve assembly **205** in an open state (or open mode). The user will then find a downhill track/trail and as they ride down the hill, they will adjust the compression adjuster **210**, e.g., by using knob **512** to adjust the adjustable valve **511**. The user will continue adjusting the adjustable valve **511** until they identify their preferred downhill vehicle performance. Once the user is satisfied with the performance of the vehicle (and shock assembly **38**) in downhill and in active or bumpy situations, the compression adjuster **210** setting is basically established for the given user.

[0160] Once the compression adjuster **210** is tuned, the user will turn the controller **39** back on which will close the active valve **521** and put the adjustable active valve assembly **205** in a closed state (or closed mode). The user will then find a climb or uphill track/trail and as they ride up the hill, they will adjust the settings of adjustable active valve assembly **205**, e.g., by using knob **527** to cause firm mode adjuster **526** to adjust the blowoff pressure of firm mode blowoff stack **525**. The user will continue using firm mode adjuster **526** to adjust the blowoff pressure of firm mode blowoff stack **525** until they find their preferred climb mode firmness. These settings would then establish the baseline compression adjuster **210** and adjustable active valve assembly **205** tune for the user. At that point, the user would then be able to make additional fine tune adjustments from their baseline depending upon the terrain being traversed, the type of ride (e.g., road, gravel, cross-country, etc.), and the like.

[0161] For example, cross country racers might crank the blowoff pressure of firm mode blowoff stack **525** close to full (or nearly full) lockout while an enduro or trail rider might want a bit of compliance so they would set the blowoff pressure of firm mode blowoff stack **525** to less than full (or nearly full) lockout.

[0162] In operation, in one embodiment, the active valve **521** would automatically switch, for example during a detected climb, to a closed position to increase firmness to avoid pedal bob. In one embodiment, a user such as, for example, an Enduro rider, may want a slightly less firm damping characteristic during the climb. As such, the Enduro rider could use knob **527** to reduce the blowoff pressure of firm mode blowoff stack **525**, and thus, change the damping characteristic of the shock (as previously and automatically selected by the controller **39** automatically closing active valve **521**) to obtain a modified damping characteristic or tune that is different than the

damping tune automatically set by the controller **39** and/or active valve **521**.

[0163] Thus, one embodiment provides a user with the capability of manually adjusting the damping characteristics or tune of shock assembly **38** (e.g., when in a firm mode) not previously found in conventional shock absorbers.

[0164] Additionally, the inclusion of both the adjustable active valve assembly **205** and the compression adjuster **210** provide a new and previously unknown “additive” component. For example, in one embodiment, compression adjuster **210** is used to achieve a desired compression setting while the adjustable active valve assembly **205** is used to achieve a further compression setting where the two components working together achieve an “additive” damping characteristic that includes a level of granularity that is not found in conventional shock absorbers.

[0165] More generally stated, embodiments of the present invention achieve a first compression mode adjustment in combination with a separate firm mode adjustment to obtain a previously unknown level of damper tune granularity. Additionally, embodiments of the present invention, utilize a manual adjustment, an automatic adjustment, and/or various combinations of manual and/or automatic adjustments to achieve the novel damping characteristics of the shock assembly **28**.

[0166] By providing this post manufacture adjustability the valve sizing (or code) choices selected by the original equipment manufacturer (OEM) and/or an aftermarket manufacturer, technician, mechanic, user, etc. become much less critical. That is, instead of having to build and live with a specific valve having a specific flow rate for a given shock assembly **38**, the flow rate for a number of the valves are now adjustable for a number of different damping characteristics after the shock assembly **38** is assembled, installed, and even while the shock assembly **38** is in use.

[0167] In one embodiment, as the active valve **521** and adjustable valve **511** comprises larger flow port(s) designed to allow the maximum fluid flow capability of the given valve assembly to pass through the parallel flow path including active valve **521** or adjustable valve **511** when they are fully open (or nearly fully-open), a pilot circuit and/or boost valve arrangement is not required in one, some, or all of the compression adjuster **210**, adjustable active valve assembly **205**, or adjustable rebound valve assembly **215**.

[0168] In one embodiment, by not requiring a boost valve arrangement, the shock assembly **38** is a smaller package with fewer parts than a shock assembly with a boost valve arrangement, which will reduce manufacture costs and speed. Moreover, shock assembly **38** as described herein will reduce clunking and provide a faster response time compared to many other shock designs.

[0169] Moreover, in one embodiment, the larger flow port design allows the maximum fluid flow capacity without creating excessively high damping including at high shaft velocities. For example, the large flow port(s) ensure that active valve **521** (or adjustable valve **511**) does not “clunk” when opening. As such, there is no need to damp (or slow down) the opening of the active valve **521** (or adjustable valve **511**) which means shock assembly **38** will be able to properly react to first strike bumps.

[0170] The examples set forth herein were presented in order to best explain, to describe particular applications, and to thereby enable those skilled in the art to make and use embodiments of the described examples. However, those skilled in the art will recognize that the foregoing description and examples have been presented for the purposes of illustration and example only. The description as set forth is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Rather, the specific features and acts described above are disclosed as example forms of implementing the Claims.

[0171] Reference throughout this document to “one embodiment,” “certain embodiments,” “an embodiment,” “various embodiments,” “some embodiments,” “various embodiments”, or similar term, means that a particular feature, structure, or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the appearances of such phrases in various places throughout this specification are not necessarily all referring to the same

embodiment. Furthermore, the particular features, structures, or characteristics of any embodiment may be combined in any suitable manner with one or more other features, structures, or characteristics of one or more other embodiments without limitation.

[0172] The foregoing Description of Embodiments is not intended to be exhaustive or to limit the embodiments to the precise form described. Instead, The examples set forth herein were presented in order to best explain, to describe particular applications, and to thereby enable those skilled in the art to make and use embodiments of the described examples. However, those skilled in the art will recognize that the foregoing description and examples have been presented for the purposes of illustration and example only. The description as set forth is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Rather, the specific features and acts described above are disclosed as example forms of implementing the Claims and their equivalents.

Claims

1. An adjustable shock assembly comprising: a damper body comprising a main chamber with a working fluid therein; an adjustable active valve assembly comprising: a first parallel flow pathway comprising an active valve; and a second parallel flow pathway comprising: a firm mode blowoff stack; a firm mode adjuster; and an externally adjustable feature coupled with said firm mode adjuster, said externally adjustable feature disposed at least partially external to said damper body, said externally adjustable feature manually manipulatable by a user of said adjustable shock assembly to adjust a blowoff pressure of said firm mode blowoff stack thereby modifying a damping characteristic of said adjustable active valve assembly; a main piston coupled with a piston shaft; and a fluid pathway, said fluid pathway fluidly coupling said main chamber with said adjustable active valve assembly which is fluidly coupled with a reservoir.
